CS & IT ENGINEERING

Digital Logic

Logic Gate

Lecture No.- 07



By - CHANDAN JHA SIR

Recap of Previous Lecture



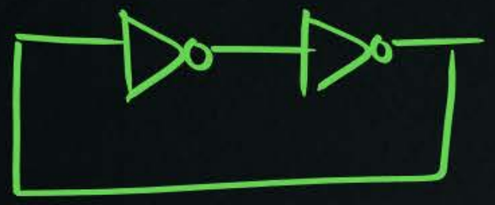
Logic GATE

Topics to be Covered



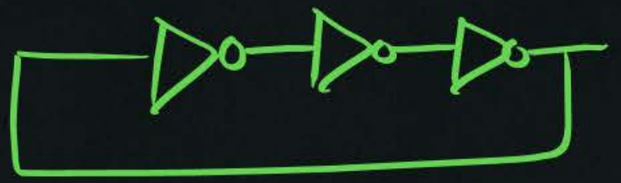
Revision & Question Practice

NOT GATE



Basic Memory

$f = 0Hz$



Astable Multivibrator

$$f = \frac{1}{2N \times \tau_{pd}}$$

AND



$A \cdot 0 = 0$

$A \cdot 1 = A$

$A \cdot \bar{A} = 0$

OR



$A + 0 = A$

$A + A = A$

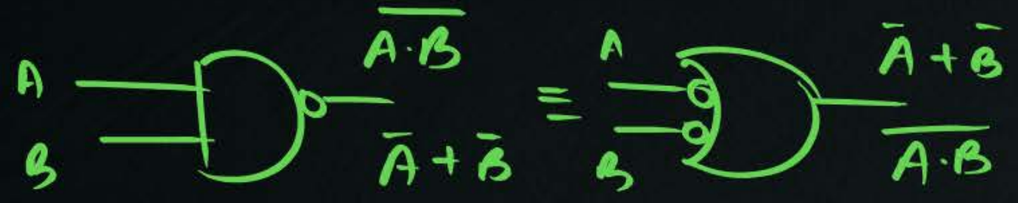
$A + 1 = 1$

$A + \bar{A} = 1$

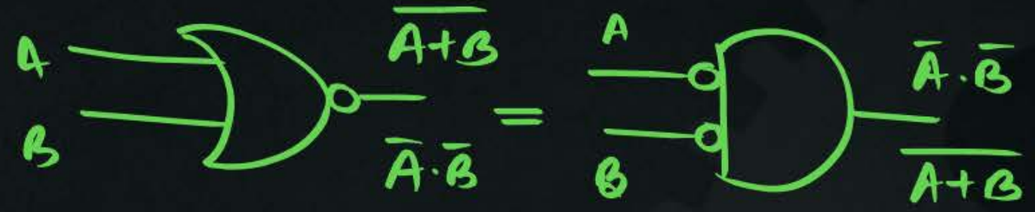
TTL

ECL

NAND



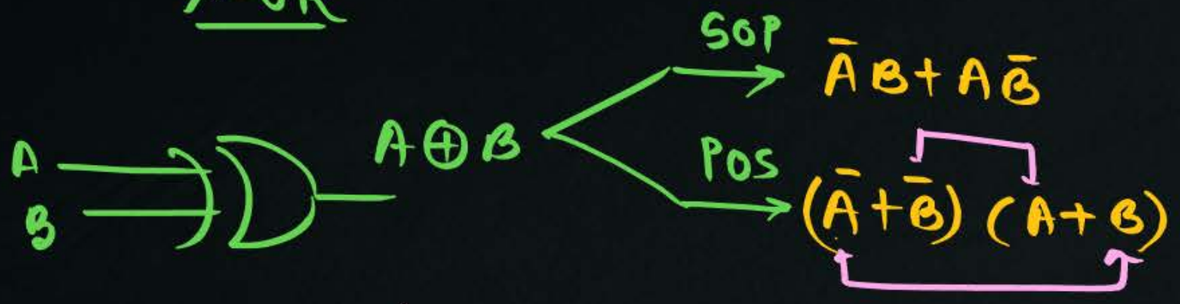
NOR



commutative Law ✓

Associative Law ✗

X-OR



A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

$A \oplus A = 0$

$A \oplus 0 = A$

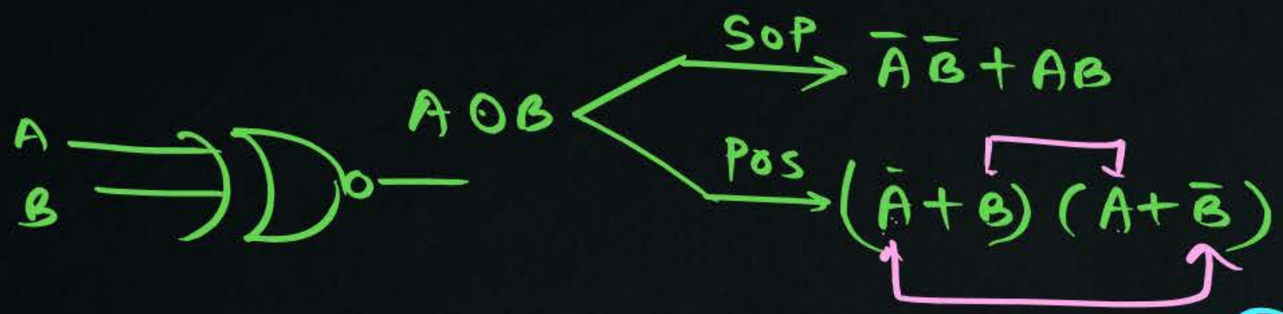
$A \oplus \bar{A} = 1$

$A \oplus 1 = \bar{A}$

$$A \oplus A \oplus \cdots \oplus A = \begin{cases} 0 & \text{Even no of A} \\ A & \text{Odd no of A} \end{cases}$$

$$A \oplus B \oplus C = \Sigma m(1,2,4,7)$$

X-NOR



A	B	A ⊙ B
0	0	1
0	1	0
1	0	0
1	1	1

$A \odot A = 1$

$A \odot 1 = A$

$A \odot \bar{A} = 0$

$A \odot 0 = \bar{A}$

$A \odot B = \overline{A \oplus B}$

$A \odot B \odot (\odot D) = \overline{A \oplus B \oplus (\oplus D}$

★ $(A \odot B) \odot C = A \oplus B \oplus C$

$\bar{A} \odot B = A \odot \bar{B} = A \oplus B = \overline{A \odot B} = \bar{A} \oplus \bar{B}$

$= A \odot \bar{B}$

$= A \oplus B$

$\bar{A} \oplus B = A \oplus \bar{B} = A \odot B = \overline{A \oplus B} = \bar{A} \odot \bar{B}$

$A \oplus \bar{B}$

$A \odot B$

① $A \cdot B \cdot C \cdot \bar{D} \cdots$

$NAND = (2n-2) + k$

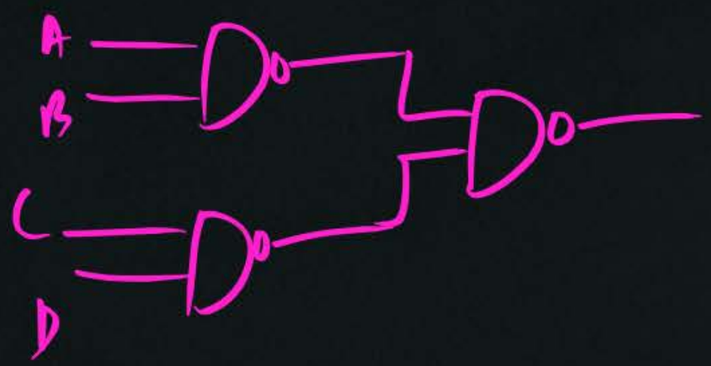
$NOR = (3n-3) - k$

② $A + \bar{B} + C + D + \cdots$

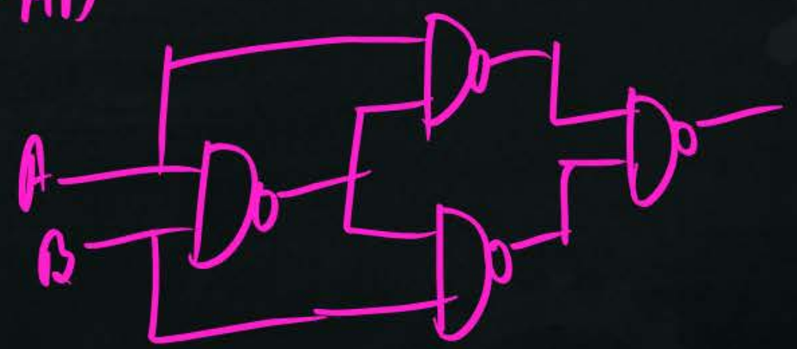
$NAND = (3n-3) - k$

$NOR = (2n-2) + k$

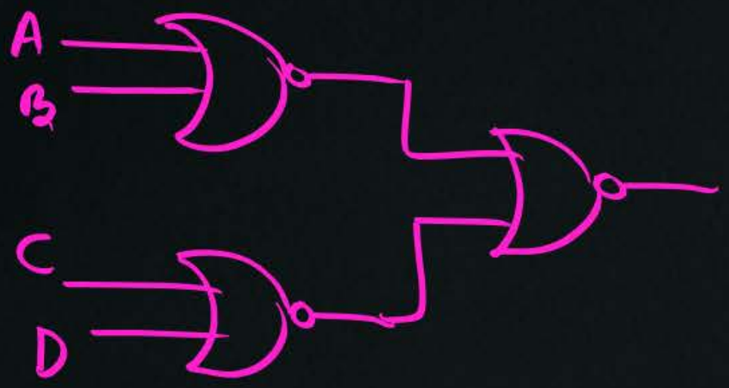
③ $AB + CD$



④ $\bar{A}B + A\bar{B}$



⑤ $(A+B)(C+D)$



NOTE

Q =



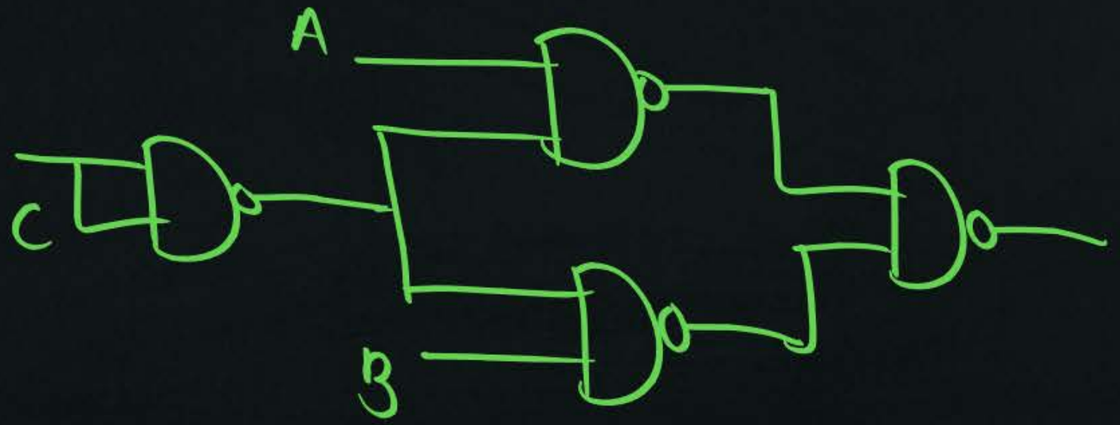
$\overline{A+B} = \bar{A}\cdot\bar{B}$

$y = \overline{\overline{A+B}+C}$

$= (A+B)\cdot\bar{C}$

$= A\cdot\bar{C} + B\bar{C}$

2 input NAND ?



(MCQ)

$A + BC = (A+B)\cdot(A+C)$

#Q. $f = (A + B)\,(A + C)\,(A + \bar{C})$ is equivalent to

$A + B C \bar{C}$

$(A+B)\cdot(A+C)\cdot(A+\bar{C})$

A $A + BC$

B $A + B\bar{C}$

$A + 0$

C 0

A

D A

(MCQ)

#Q. A logic function is given as:

$f(A, B, C, D) = [A + B\bar{C}D + \bar{B}CD + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}]\,B\bar{C}$

is equivalent to

$A B\bar{C} + B\bar{C}D + \bar{A}B\bar{C}$

$B\bar{C}[A + D + \bar{A}]$

$B\bar{C}[1 + D] = B\bar{C}$

A $A\bar{B}CD$

B $B\bar{C}$ ✓

C $A\bar{B} + B\bar{C} + CD$

D $AB\bar{C}D$

(NAT)

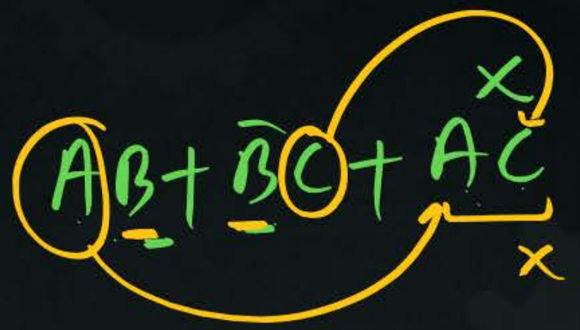
#Q. If we have 4-variables in a logic function, the number of non-dual logical functions possible are_____.

#

Distribution Theorem :→

$$A+BC=(A+B)(A+C)$$

$$A+BCD=(A+B)(A+C)(A+D)$$

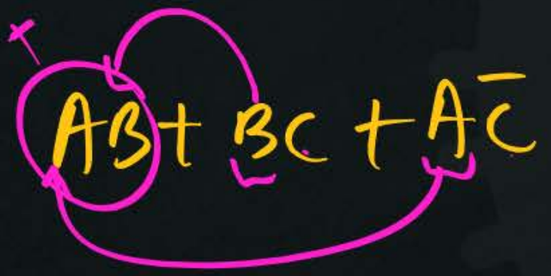


$AB+\bar{B}C$ ✓

Concensus Theorem / Redundent Theorem

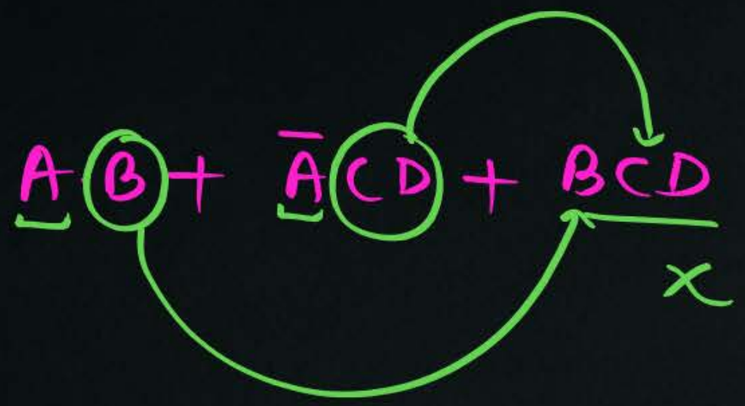


$=AB+\bar{A}C$



$BC+A\bar{C}$

$=AB$



$AB + \bar{A}CD$

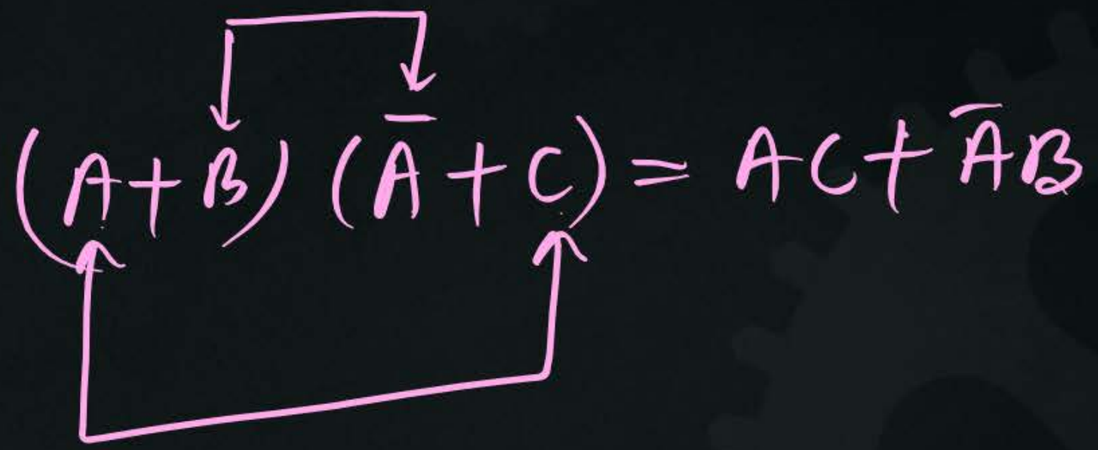
Transpose Theorem

$(A+B)(\bar{A}+C)$

$A\cdot\bar{A}+AC+\bar{A}B+BC$

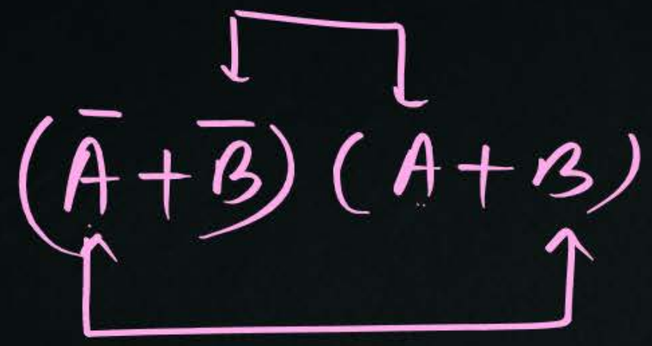
$AC+\bar{A}B+BC$

$AC+\bar{A}B$

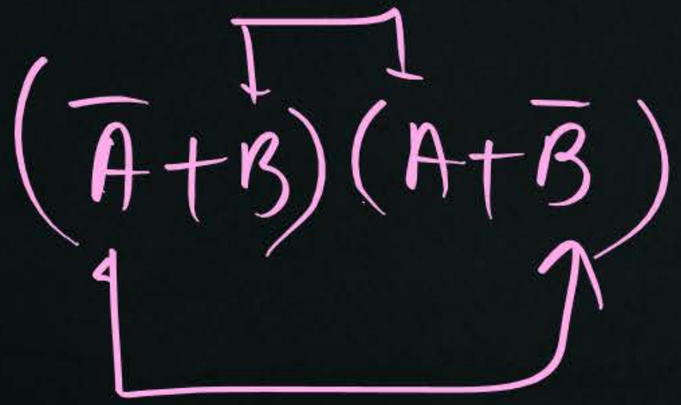


$(A+B)(A+C)=A+BC$

$(A+B)(\bar{A}+C)=AC+\bar{A}B$



$\bar{A}B+A\bar{B}=A\oplus B$



$\bar{A}\bar{B}+AB=A\odot B$

$A+B\bar{B}=A+0=A$

$(A+B)(A+\bar{B})$

$A\cdot A+A\bar{B}+AB+B\bar{B}$

$A+A\bar{B}+AB$

$A(1+\bar{B}+B)=A$

D- Morgan's Law

$$\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$$

$$\overline{A+B+C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

$$A + BC = (A+B)(A+C)$$

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

$$(A+B)(\bar{A}+C) = AC + \bar{A}B$$

$$\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$$

$$\overline{A+B+C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

(MCQ)

#Q. A logic function
$f(A, B, C) = (A + B)(\bar{B} + C)(A + C)$, then $\bar{f}$ will be equal to

#

A $AB + \bar{B}C$

B $\bar{A}\bar{B} + B\bar{C}$

C $\bar{A}\bar{B} + \bar{A}\bar{C}$

D $AB + AC$

$(A+B)(A+C)\cdot(\bar{B}+C)$

$(A+BC)(\bar{B}+C)$

$A\bar{B}+BC\cdot\bar{B}+AC+BC\cdot C$

$A\bar{B}+0+AC+BC$

$A\bar{B}+AC+BC$

$A\bar{B}+BC$

$f = A\bar{B}+BC$

$\bar{f} = \overline{A\bar{B}+BC}$

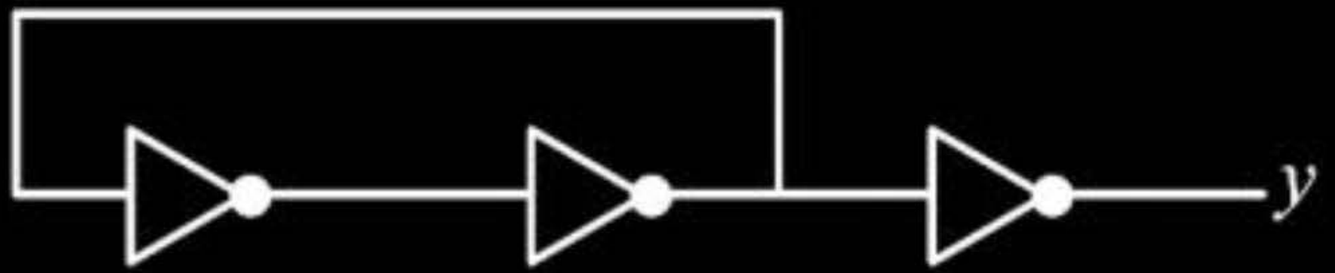
$= \overline{A\bar{B}}\cdot\overline{BC}$

$= (\bar{A}+B)\cdot(\bar{B}+\bar{C})$

$= (B+\bar{A})(\bar{B}+\bar{C}) = B\bar{C}+\bar{A}\bar{B}$

(MCQ)

#Q. A digital circuit is designed as shown:
This circuit works as



A Astable multivibrator as odd number of NOT gates are used.

B Bistable multivibrator

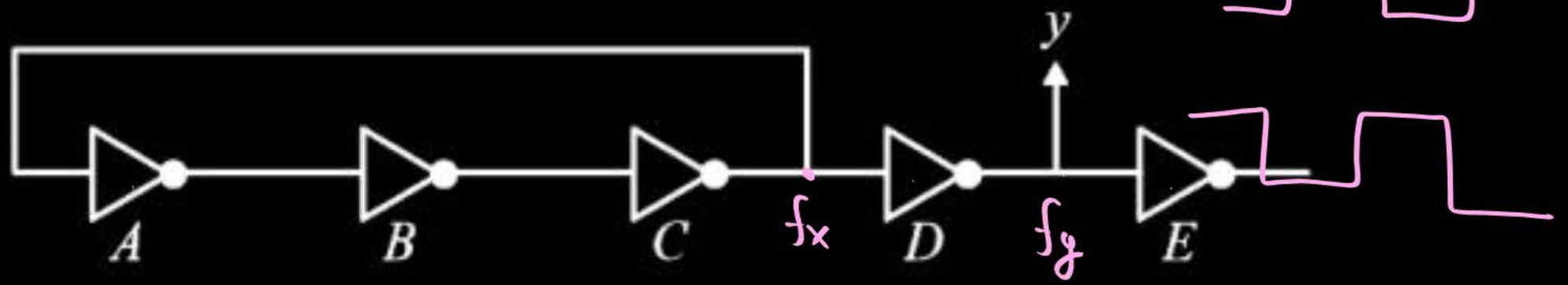
C Monostable multivibrator

D None of these

(NAT)

$f_x = f_y$

#Q. An astable multivibrator circuit is designed as shown:



NOT gates A, B & C are of logic family TTL with delay t_d = 20/3 nsec of each gate and gates D & E are of logic family ECL with delay t'_d =10/3 nsec of each gate, then frequency of the waveform at output y is _____ MHz.

$$f_x = \frac{1}{2 \times 3 \times \tau_{pd}} = \frac{1}{2 \times 3 \times \frac{20}{3} \times 10^{-9}} \text{ s}^{-1} = \frac{10^9}{40} \quad H_z = \frac{1000 \times 10^6}{40}$$

$$= 25 \text{ MHz}$$

(MCQ)

#Q. Logical function $f(A, B, C, D) = AB + \bar{A}CD + \bar{B}CD$ is equivalent to

$= AB + CD(\bar{A} + \bar{B})$

$= AB + CD \cdot \overline{AB}$

$AB = X$

$= X + CD \cdot \bar{X}$

$= (X + CD)(X + \bar{X})$

$= X + CD = AB + CD$

A $AB + \bar{B}C$

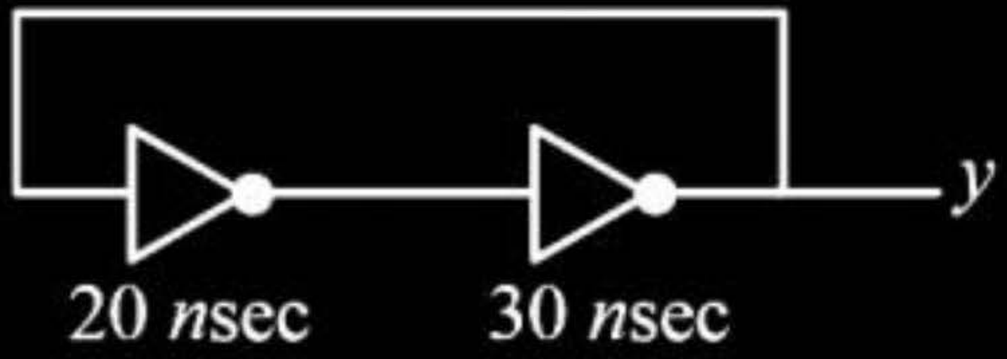
B $AB + CD$

C $\bar{A}C + \bar{B}C$

D $AB + B\bar{C}$

(MCQ)

#Q. A logical circuit is designed as shown:
Output y is



A A bistable multivibrator's output

B A square waveform with $f = 10$ MHz

C A square waveform with $f = 25$ MHz

D None of the above.

(MCQ)

?

#Q. Which of the following is true?

$\overline{A \oplus B} = A \odot B$

$A \odot B$ — $\bar{A}\bar{B} + AB$ — $(\bar{A}+B)(A+\bar{B})$

A $\overline{\bar{A}B + A\bar{B}} = (\bar{A}+\bar{B})(A+B)$

B $\overline{ABC\bar{D}} = \bar{A}+\bar{B}+\bar{C}+\bar{D}$

C $\overline{\overline{A\bar{B}}.C} = (A+\bar{C})(\bar{B}+\bar{C})$

D None of these

$\overline{\overline{A\bar{B}}.C}$

$A\bar{B} + \bar{C} = (A+\bar{C})(\bar{B}+\bar{C})$

$$\overline{A B \overline{C D}} = \overline{A B} + C D$$

$$= \bar{A} + \bar{B} + C D$$

(MCQ)

#Q. Which of the following is true?

A We can use '1' as enable input for OR gate ✗

B We can use '0' as enable input for AND gate ✗

C '0' as well as '1' can be used as enable input for XNOR gate ✓

D None of the these

(MCQ)

#Q. Which of the following relation is true?

$A \oplus \bar{B} = A \odot B = \bar{A} \oplus B$

A. $A \oplus \bar{B} = \bar{A} \odot B$ ✗

B. $\overline{A \oplus \bar{B}} = A \odot B$ ✗

C. $\overline{\bar{A} \odot \bar{B}} = A \oplus B$ ✓

D. $\overline{\bar{A} \oplus \bar{B}} = A \oplus B$ ✗

$\overline{A \oplus \bar{B}}$

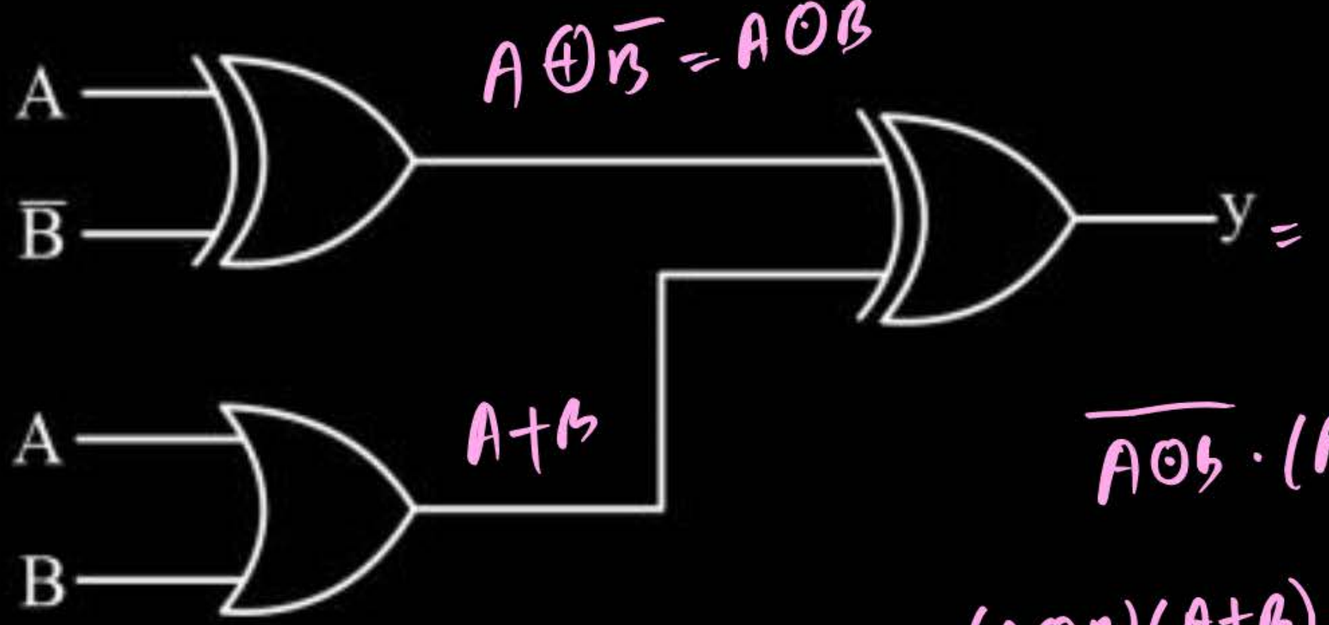
$\overline{A \odot B} = A \oplus B$

$\overline{\bar{A} \odot \bar{B}}$

$\overline{A \odot B} = A \oplus B$

(MCQ)

#Q. A logical circuit is as given below:
Output y will be



$A \oplus \bar{B} = A \odot B$

$y = (A \odot B) \oplus (A + B)$

$\overline{A \odot B} \cdot (A + B) + (A \odot B)\overline{A+B}$

$(A \oplus B)(A + B) + (A \odot B)(\bar{A}\bar{B})$

$(\bar{A}B + A\bar{B})(A + B) + (\bar{A}\bar{B} + AB)\bar{A}$

A $\bar{A} + B$

B $\bar{A} + \bar{B}$

C $A\bar{B}$

D $A + B$

$$(\bar{A}B + A\bar{B})(A+B) + (\bar{A}\bar{B} + AB)\cdot \bar{A}\bar{B}$$

$$\cancel{\bar{A}B\cdot A}^{0} + A\bar{B}\cdot A + \bar{A}B\cdot B + \cancel{AB\cdot B}^{0} + \bar{A}\bar{B}\cdot\bar{A}\bar{B} + \cancel{AB\cdot\bar{A}\bar{B}}^{0}$$

$$A\bar{B} + \bar{A}B + \bar{A}\bar{B}$$

$$A\bar{B} + \bar{A}(B+\bar{B})$$

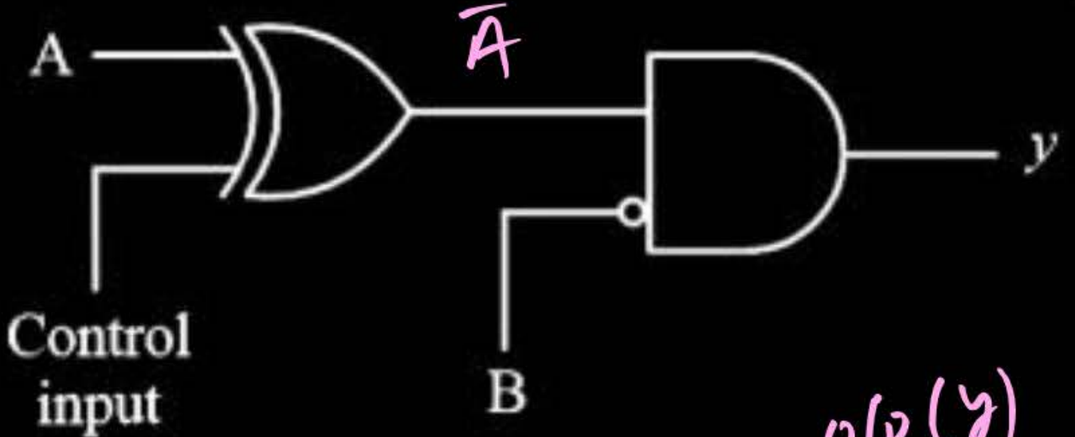
$$\bar{A} + A\bar{B}$$

$$(\bar{A}+A)\cdot(\bar{A}+\bar{B})$$

$$\bar{A}+\bar{B}$$

(MCQ)

#Q. A logic circuit is as given below:
Which of the following is true?



Control input | O/P (y)
0 | $A\bar{B}$
1 |

$\bar{A} \cdot \bar{B} = \overline{A+B}$

A Output y is $\bar{A}B$ if control input = 0 ✗

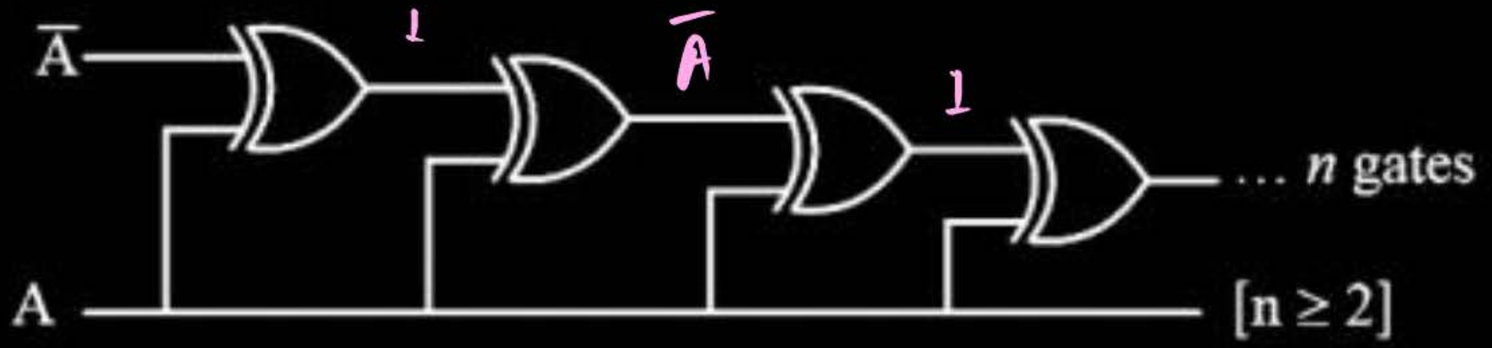
B Output y is $\overline{A+B}$ if control input = 1 ✓

C Output y is $\overline{A \cdot B}$ if control input = 0 ✓

D Output y is $\overline{\bar{A} \cdot B}$ if control input = 1 ✗

(MCQ)

#Q. A logic circuit is as given below:
Which of the following is true?



A Output is $\bar{A}$ if n is even ✓

B Output is A if n is even

C Output is $\bar{A}$ if n is odd

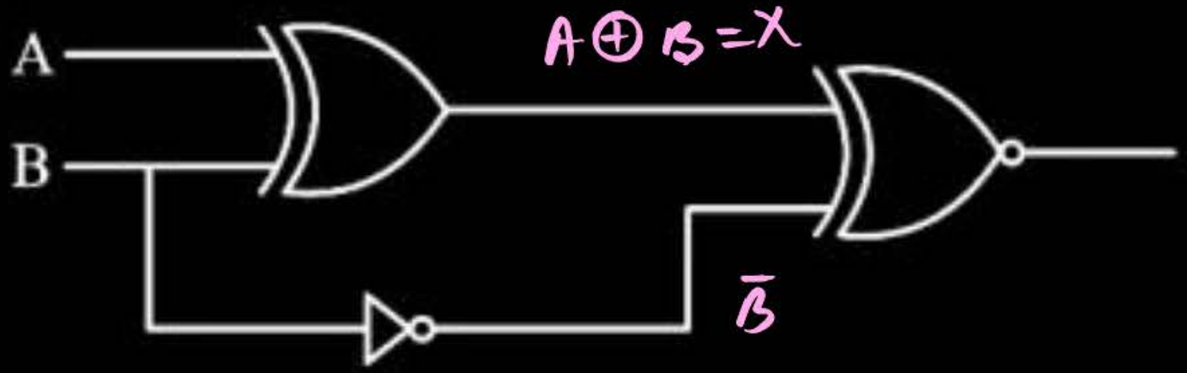
D Output is A if n is odd

n = odd → 1

n = even → $\bar{A}$

(MCQ)

#Q. A logical circuit is as given below:
Output y is



A A

B $\bar{B}$

C $\bar{A}$

D B

(NAT)

#Q. A logical expression is given as:

$$f(A,B,C,D) = \bar{A} + AB[ABC + \bar{B}C + AB\bar{C} + C\bar{D}]$$

The minimum number of 2-input NAND gate required to implement above logic function will be ____ .

$\bar{A} + AB + AB\bar{C} + ABC\bar{D}$

$\bar{A} + AB[1 + \bar{C} + C\bar{D}]$

$\bar{A} + AB = \bar{A} + B$

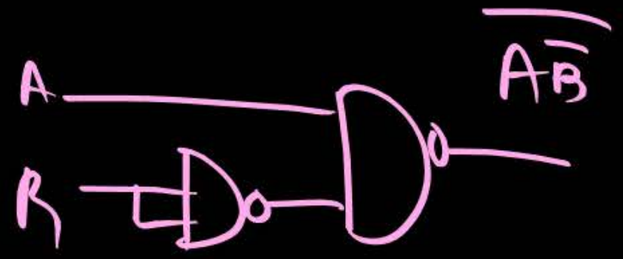
$n=2$
$k=1$

$\bar{A} + B = \overline{A \cdot \bar{B}}$

$(3n-3)-k$

$(6-3)-1$

$3-1=2$



(NAT)

#Q. A logical expression is given as:

$f(A, B, C) = (\bar{A} + B)\ (A + \bar{B})$, the minimum number of 2-input NAND gate required to implement above logical function is _____ .

$\bar{A}\bar{B} + AB = A \odot B$

2

(NAT)

#Q. A logical expression is given as:

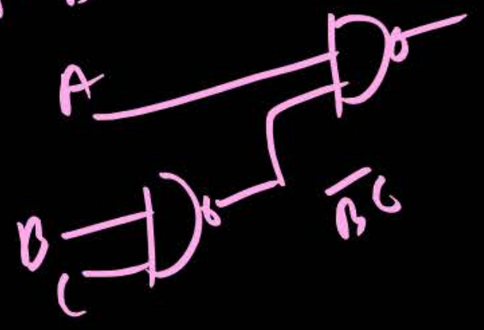
$f(A, B, C) = \bar{A} + ABC$, then minimum number of 2-input NAND gate required to implement above logical function is _____.

$(\bar{A}+A)(\bar{A}+BC)$

$\bar{A}+BC$

$\overline{A \cdot \overline{BC}}$

(2) Ans



(NAT)

#Q. A logical function is given as:

$f(A, B) = A \oplus A\bar{B}$, If we implement this logical function using 2-input NAND gate, the minimum number of NAND gate required is _____ .

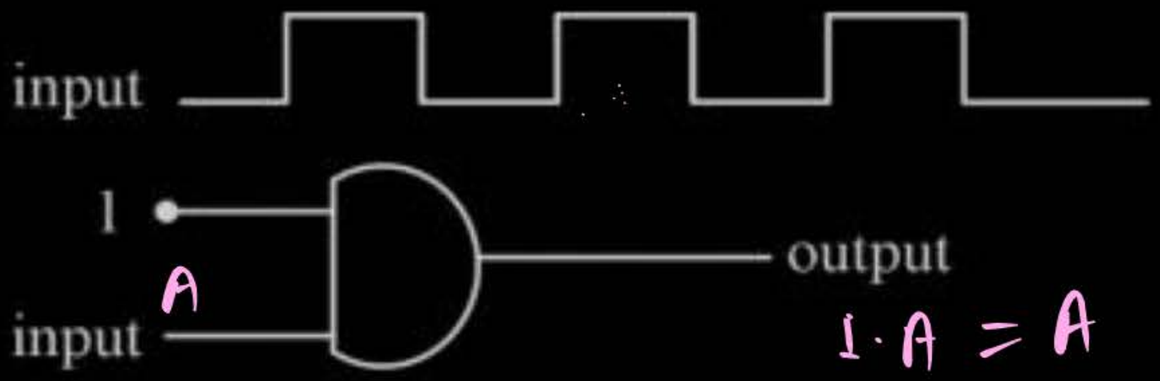
2

$A \oplus A\bar{B} = \bar{A}A\bar{B} + A \cdot \overline{A\bar{B}}$

$A \cdot (\bar{A} + B) = AB$



#Q. The input wave form is given



Draw output wave form

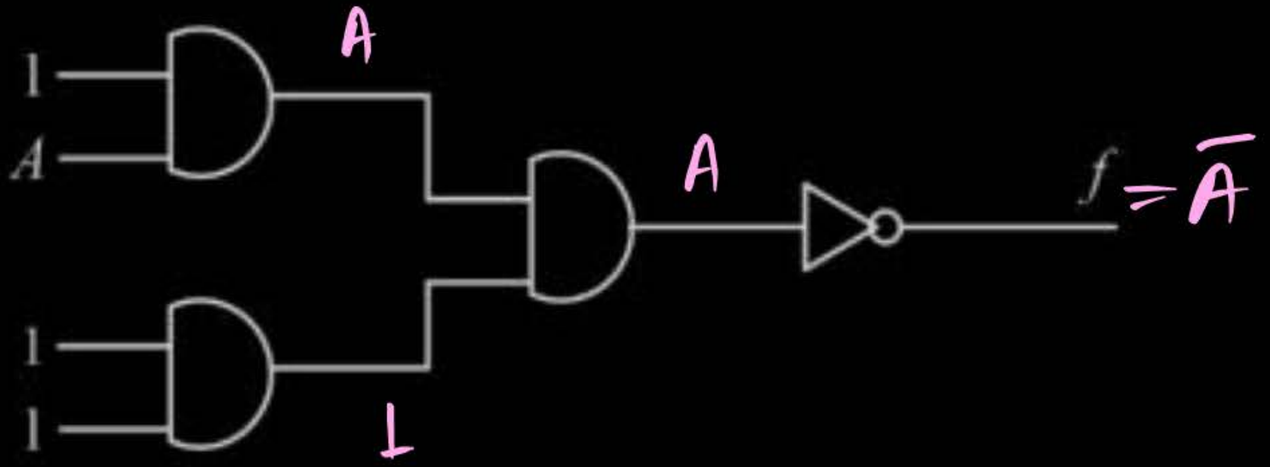
A 

B 0

C 1

D 

#Q. Find value of f ?



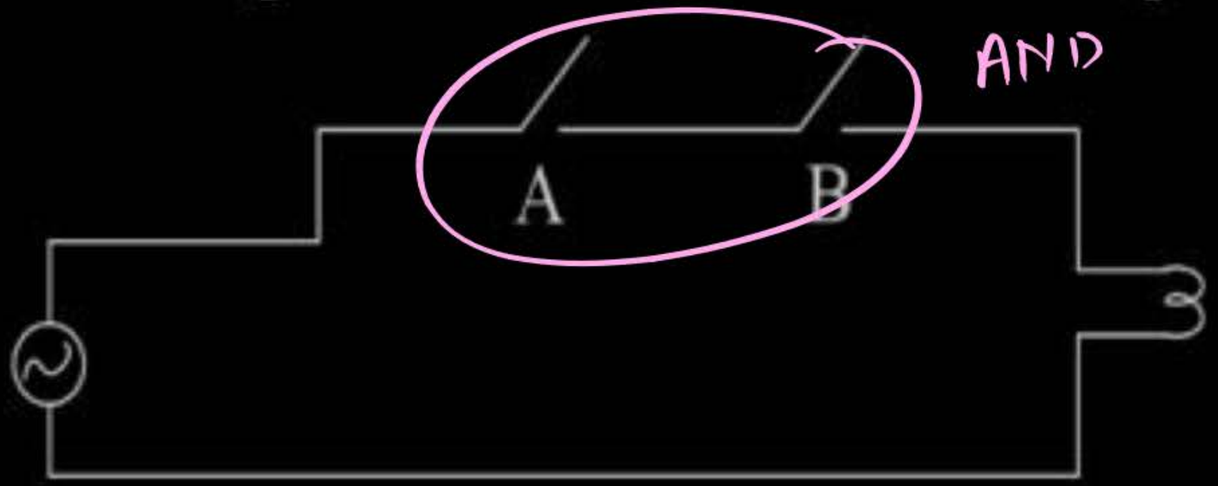
A 1

B 0

C A

D $\overline{A}$

#Q. Bulb will glow when A and B are respectively.



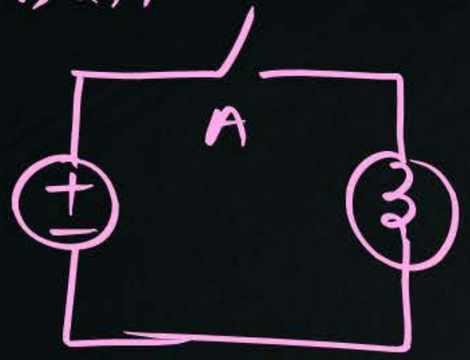
A 00

B 01

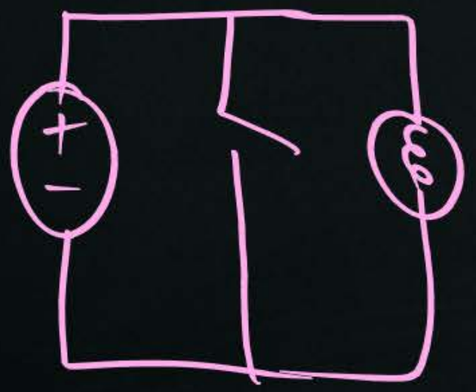
C 10

D 11

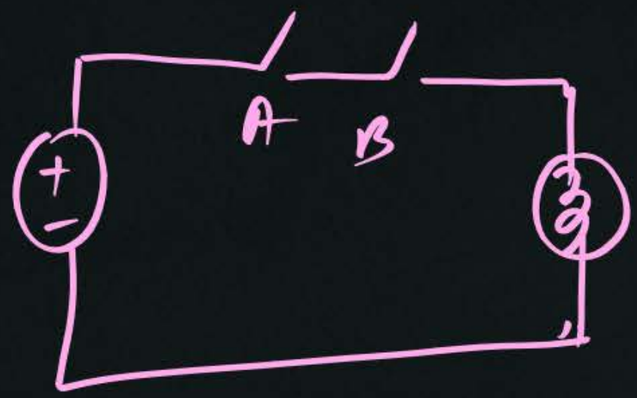
① Buffer



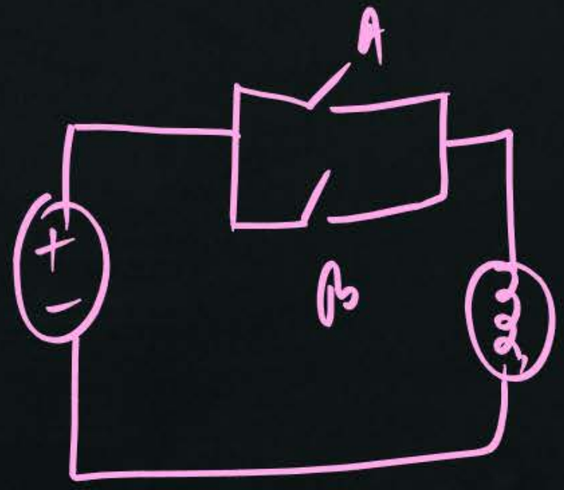
② NOT



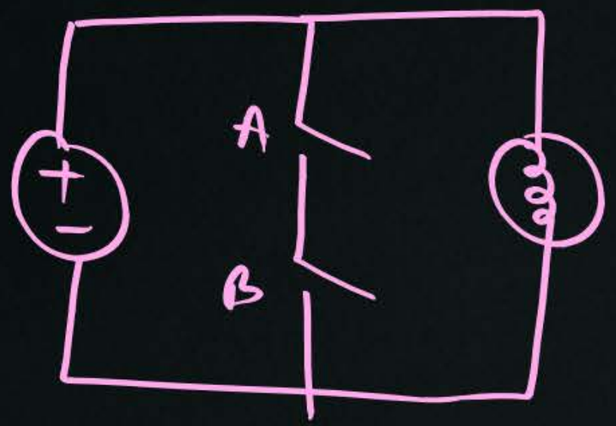
③ AND



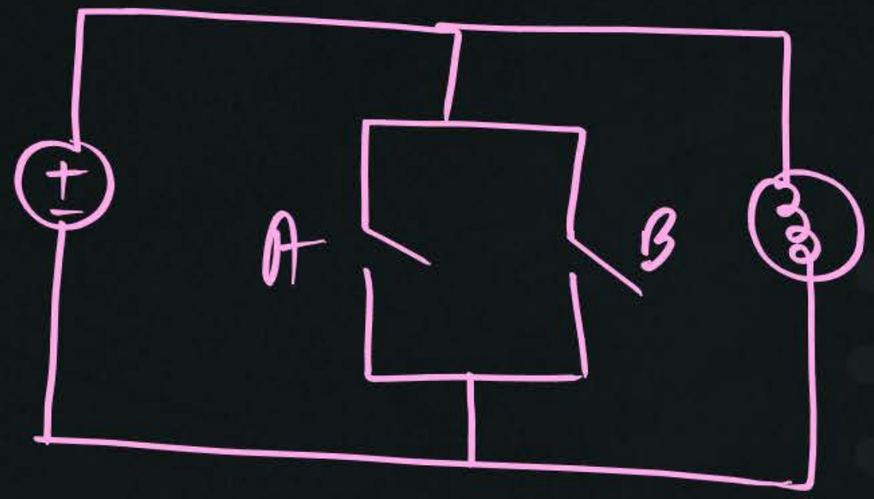
④ OR



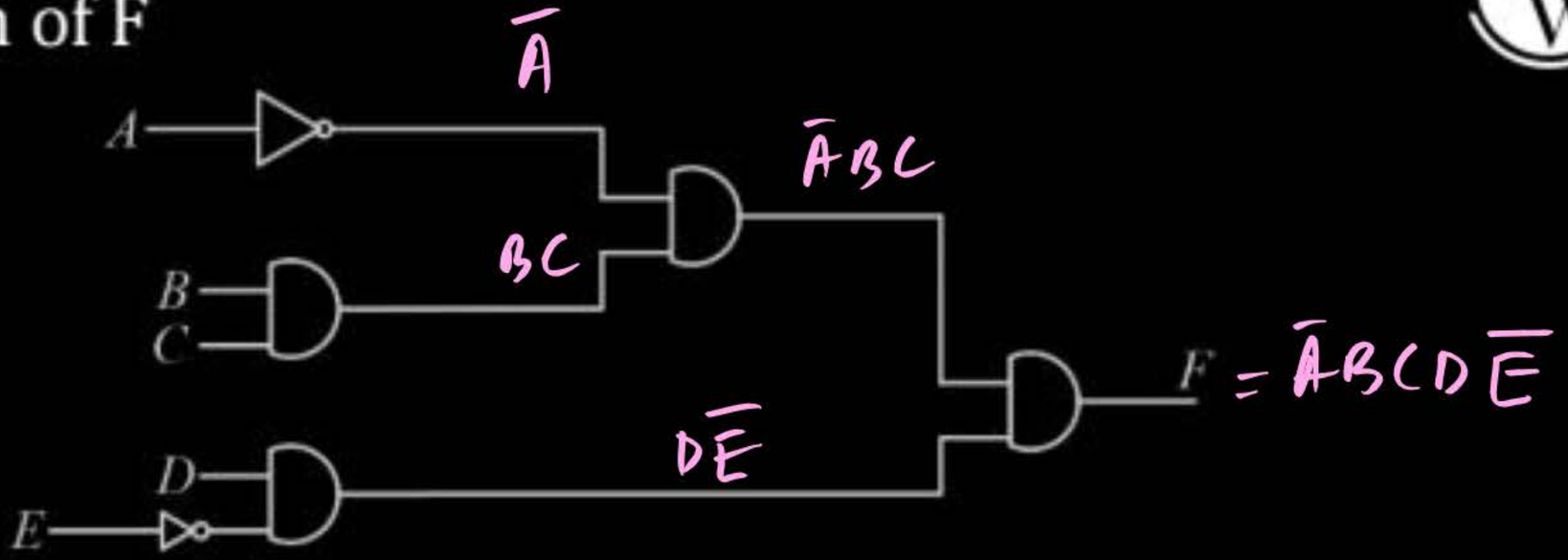
NAND



NOR



#Q. Find expression of F



A $\bar{A}BC + D\bar{E}$

B $\bar{A}BCD\bar{E}$

C $(\bar{A} + BC)D\bar{E}$

D $\bar{A}BC(D + \bar{E})$

#Q. The inverter is _____ gate

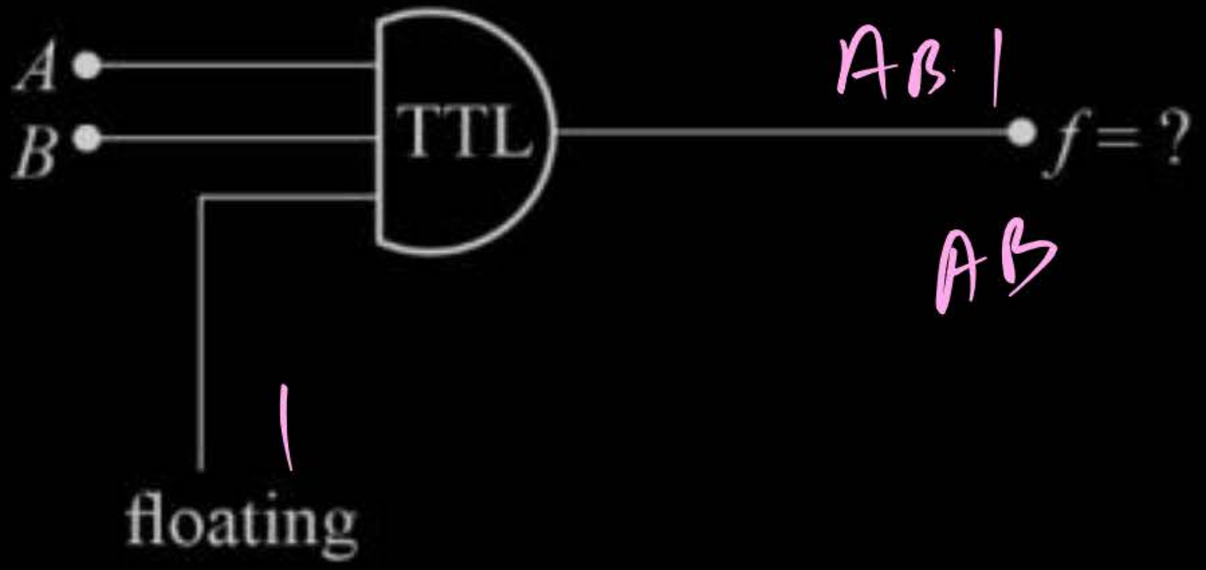
A AND

B OR

C NOT

D none of the above

#Q.



Find expression of f ?

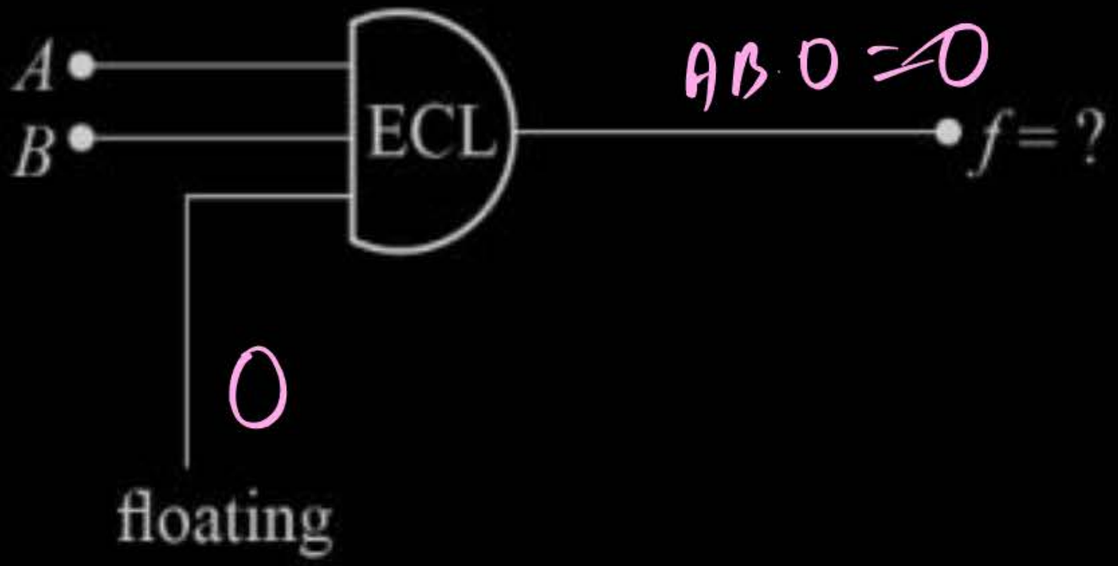
A AB

B 1

C A + B

D $\overline{AB}$

#Q. Find expression of f ?



A 0

B 1

C AB

D $\overline{AB}$

#Q. Find output frequency, if the propagation delay of NOT gate is 2 n sec.



N = 1

$$f = \frac{1}{2N\tau_{pd}}$$

$$= \frac{1}{2 \times 2 \times 10^{-9}} \text{ Hz}$$

$$= \frac{1000}{4} \text{ MHz}$$

$$= 250 \text{ MHz}$$

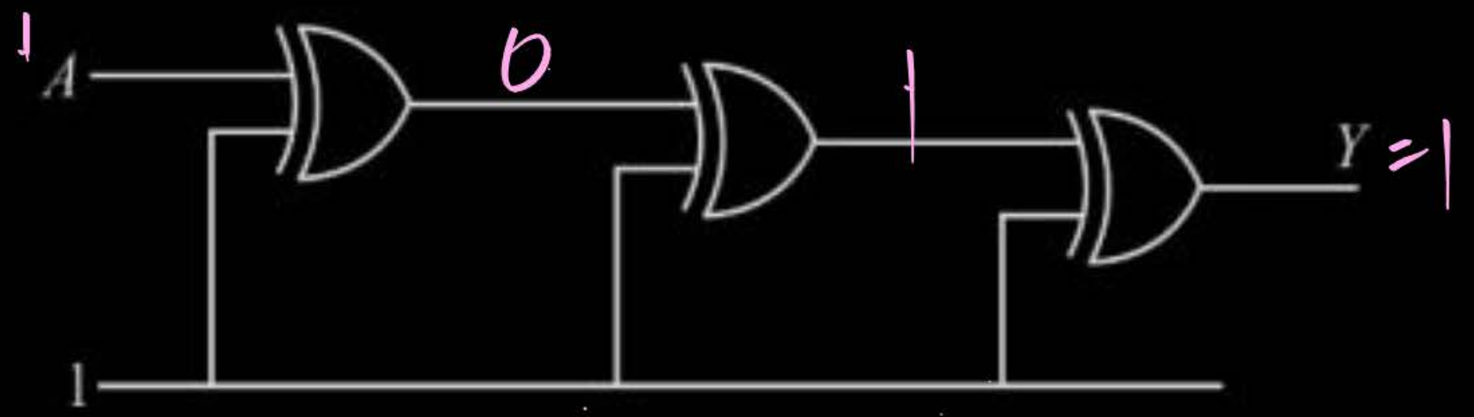
A 125 MHz

B 500 MHz

C 250 MHz

D none of the above

#Q. The initial output of the following circuit is 1. If we apply 010101 at input A (first bit is zero), then what is the bit pattern generated at the output Y.



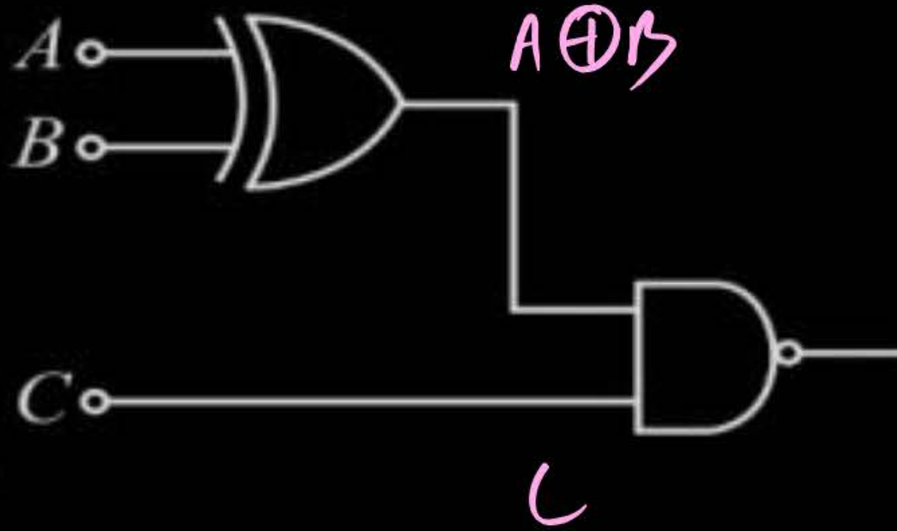
A 010101

B 101010

C remains at 0

D remains at '1'

#Q. The Boolean expression of the output of the logic circuit shown in figure is



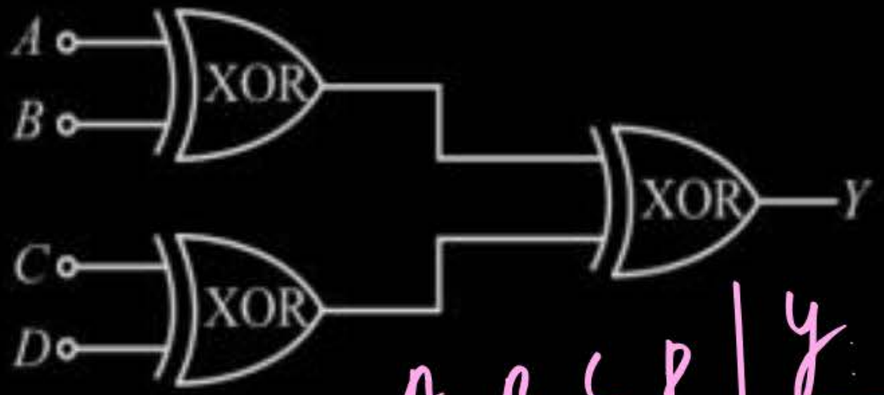
A $Y = \bar{A}\bar{B} + AB + \bar{C}$

B $Y = \bar{A}\bar{B} + AB + C$

C $Y = \bar{A}B + \bar{A}B + C$

D $Y = \bar{A}B + \bar{A}\bar{B} + C$

#Q. A, B, C and D are input, and Y is the output bit in the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?



A	B	C	D	Y	S = A+B+C+D
0	0	0	0	0	0
0	0	0	1	1	1
0	0	1	0	1	1

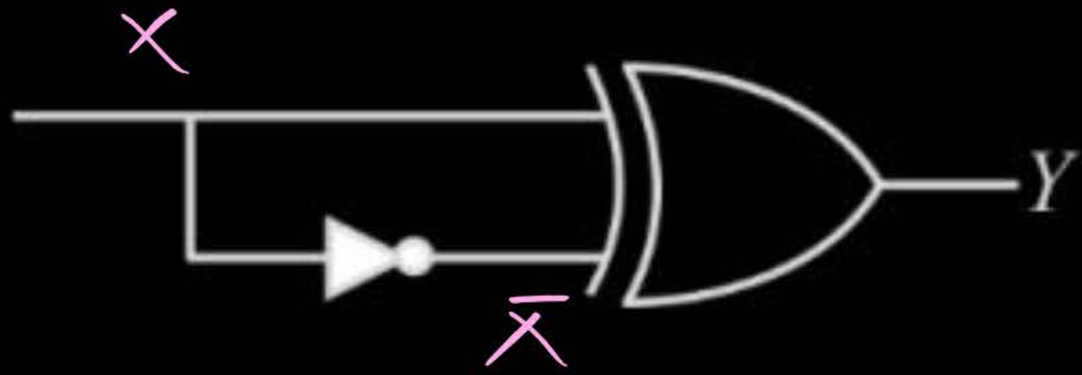
A S is always either zero or odd.

B S is always either zero or even.

C S = 1 only if the sum of A, B, C and D is even.

D S = 1 only if the sum of A, B, C and D is odd.

#Q. The output Y of the logic circuit given below is



$= X \oplus \bar{X}$

$= 1$

A 1

B 0

C X

D $\overline{X}$

#Q. A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

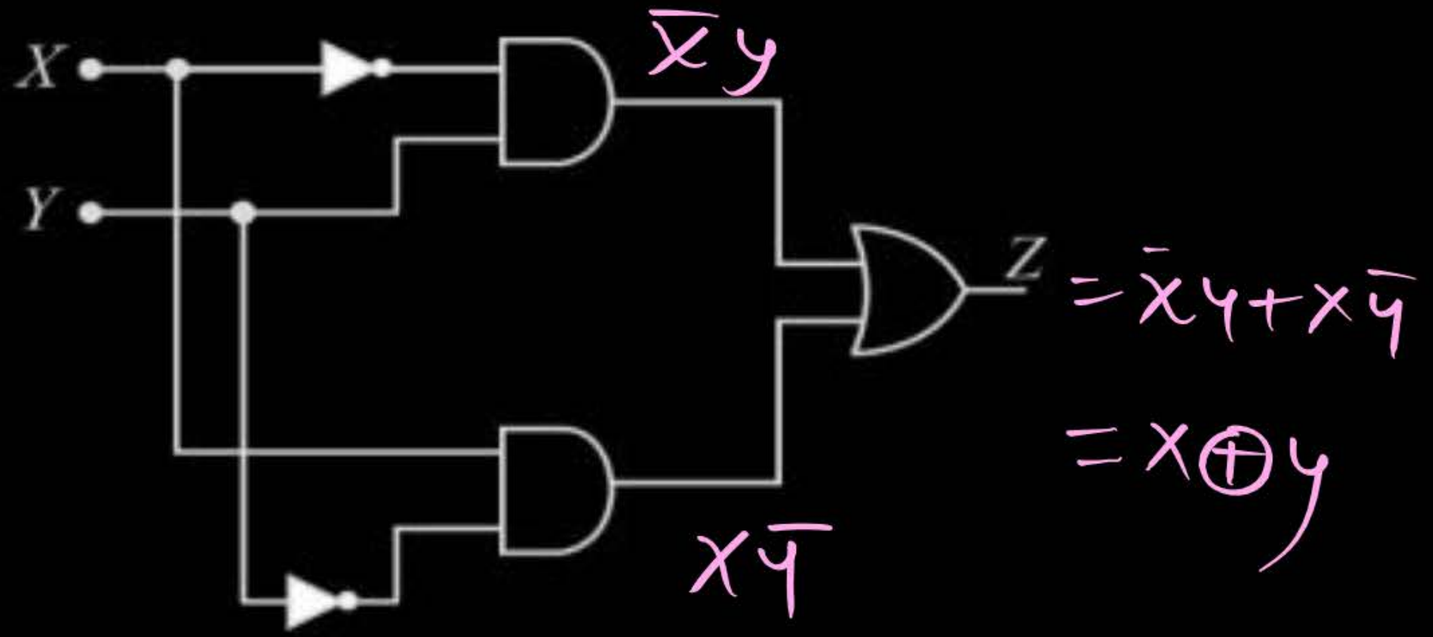
A an AND gate

B an OR gate

C an XOR gate

D a NAND gate

#Q. In the circuit shown below, X and Y are digital inputs, and Z is a digital output. The equivalent circuit is



A XNOR gate

B NOR gate

C NAND gate

D XOR gate

#Q. Which one of the following gate is also known as equivalence gate?

A EX-OR

B AND

C EX-NOR

D NOR

#Q. The logic function $f = \overline{xy + \bar{x}\,\bar{y}}$ is equal to

A EX-NOR

B NAND

C EX-OR

D NOR

#Q. The minimum number of 2 input NOR gates required to implement a 2 input XOR gate is_____

5

#Q. The minimum number of 2 input NAND gates required to implement a 2 input EX-NOR gates is

A 4

B 5

C 3

D 6





2 Min Summary

Thank you

GW

Soldiers !

